SaxonC - Bug #7041

SaxonApiException after X Saxon\XsltExecutable::transformFileToString

2026-03-10 12:27 - Benjamin DIELEMAN

Status:	Resolved	**Beginn:**	2026-03-10
Priorität:	Normal	**Abgabedatum:**	
Zugewiesen an:	O'Neil Delpratt	**% erledigt:**	100%
Kategorie:	PHP API	**Geschätzter Aufwand:**	0:00 Stunde
Sprint/Meilenstein:	12.9		
Applies to branch:	12, trunk	**Fixed in version:**	
Fix Committed on Branch:	12	**SaxonC Languages:**	All
Fixed in Maintenance Release:		**SaxonC Platforms:**	All
Found in version:		**SaxonC Architecture:**	

Beschreibung

Hello, When we use the same XsltExecutable for multiple transformation, a C++ uncatchable exception can be thrown by the library :

```
terminate called after throwing an instance of 'SaxonApiException'
  what():  std::exception
```

XsltExecutable seems to no keeping reference to SaxonProcessor et Xslt30Processor.

Here is how to reproduce the bug:

```
$executable = $this->getXsltExecutable($stylesheet);
for($i=0; $i<5000; $i++) {
    $tmpFileWarning = tempnam(sys_get_temp_dir(), 'schematron-validator-warning');
    $executable->setSaveXslMessage(true, $tmpFileWarning);
    $xmlResult = $executable->transformFileToString($filePath);
    unlink($tmpFileWarning);
    unset($tmpFileWarning);
}
```

We don't have the issue, if we build the executable on the same method, or if we keep a reference to SaxonProcessor et Xslt30Processor in class properties.

```
$saxon = new \Saxon\SaxonProcessor();
$xslt30Processor = $saxon->newXslt30Processor();
$executable = $xslt30Processor->compileFromFile($stylesheet);
for($i=0; $i<5000; $i++) {
    $tmpFileWarning = tempnam(sys_get_temp_dir(), 'schematron-validator-warning');
    $executable->setSaveXslMessage(true, $tmpFileWarning);
    $xmlResult = $executable->transformFileToString($filePath);
    unlink($tmpFileWarning);
    unset($tmpFileWarning);
}
```

Best regards.

Historie

#1 - 2026-03-10 13:28 - O'Neil Delpratt

Thanks for reporting the issue you have found. I will try to reproduce the error based on the snippets you have provided.

Please can you check what happens if you call $executable->exceptionClear() at the end inside the for loop?

#2 - 2026-03-10 13:56 - Benjamin DIELEMAN

Hello, I tried a lot of things, $executable->exceptionClear() didn't fix this issue.

As a temporary fix, I keep a reference to $saxon and `$xslt30Processor in class properties, so GC won't remove the object.

#3 - 2026-03-10 14:39 - O'Neil Delpratt

Ok thanks for trying that. The other issue I see is the setting of the XslMessage (i.e. $executable->setSaveXslMessage(true, $tmpFileWarning);). Have you tried clearing the XslMessage listener? i.e. In the for loop at the end $executable->clearXslMessages()

Do you have a minimised full code that I can run at my end please?

#4 - 2026-03-10 14:43 - O'Neil Delpratt

Also does this code need to be in the for loop:

```
    $tmpFileWarning = tempnam(sys_get_temp_dir(), 'schematron-validator-warning');
    $executable->setSaveXslMessage(true, $tmpFileWarning);
```

If its always the same data you could move this before the loop.

#5 - 2026-03-10 15:00 - Benjamin DIELEMAN

I put it in the loop to simulate a real XSLT on a file that changes with each call.

#6 - 2026-03-10 15:10 - Benjamin DIELEMAN

- Dateianhang PEPPOL-EN16931-UBL.xsl wurde hinzugefügt

O'Neil Delpratt wrote in #note-3:

> Ok thanks for trying that. The other issue I see is the setting of the XslMessage (i.e. $executable->setSaveXslMessage(true, $tmpFileWarning);). Have you tried clearing the XslMessage listener? i.e. In the for loop at the end $executable->clearXslMessages()

ahaha ! Yes, that was one of my theories :). I tried clearXslMessages(), but it doesn't fix the problem.

O'Neil Delpratt wrote in #note-3:

> Do you have a minimised full code that I can run at my end please?

```
public function getXsltExecutable(string $stylesheet) : \Saxon\XsltExecutable
    {
        $saxon = new \Saxon\SaxonProcessor();
        $xslt30Processor = $saxon->newXslt30Processor();
        return $xslt30Processor->compileFromFile($stylesheet);
    }
```

Enclosed you'll find a stylesheet. I don't think XML is important; you should be able to use any one at random ?

#7 - 2026-03-11 12:35 - O'Neil Delpratt

- Status wurde von New zu In Progress geändert

- Applies to branch 12, trunk wurde hinzugefügt

I have managed to track down the issue. There is a bug here. When you create the XsltExecutable you should not need to keep around the Xslt30Processor. But in the XsltExecutable.transformFileToString() we have an unnecessary internal check for the Xslt30Processor. When the PHP GC is triggered this check for the Xslt30Processor causes a crash because it has been destroyed.

We also need to check the other executable methods in XsltExecutable for the same problem.

As the user mentioned the workaround is to keep a global reference to the Xslt30Processor object. I don't think we need to do the same for the SaxonProcessor object.

#8 - 2026-03-11 15:34 - O'Neil Delpratt

I was wrong about what I thought was the cause of this issue.

I managed to reproduce the exception. With debug option switched on the underlying problem is as follows:

```
net.sf.saxon.option.cpp.SaxonCMessageListener cannot be cast to net.sf.saxon.s9api.Processor
java.lang.ClassCastException: net.sf.saxon.option.cpp.SaxonCMessageListener cannot be cast to net.sf.saxon.s9a
pi.Processor
 at net.sf.saxon.option.cpp.SaxonCMessageListener.createSaxonCMessageListener(SaxonCMessageListener.java:79)
```

#9 - 2026-03-11 17:21 - O'Neil Delpratt

Now that I am able to reproduce the exception, we have all the clues needed to identify the cause. The suspicion mentioned in Comment #7 was not far off. The call to setSaveXslMessage creates a MessageListener in the underlying Java code. To create this object, a SaxonProcessor is required. However, the SaxonProcessor is no longer in scope and has been garbage collected by PHP.

#10 - 2026-03-11 17:49 - O'Neil Delpratt

The other processors also internally keep a reference the SaxonProcessor. This needs to change.

#11 - 2026-03-16 11:07 - O'Neil Delpratt

- Status wurde von In Progress zu Resolved geändert

- % erledigt wurde von 0 zu 100 geändert

- Fix Committed on Branch 12 wurde hinzugefügt

- SaxonC Languages All wurde hinzugefügt

- SaxonC Languages PHP wurde gelöscht

- SaxonC Platforms All wurde hinzugefügt

- SaxonC Platforms Linux wurde gelöscht

Bug fixed in all the processors. In the Processors (Xath, XSLT, XQuery and Schema) we no longer keep an internal reference to the C++ SaxonProcessor object which created them. Instead we create a new internal Java object handle to the SaxonProcessor object.

Dateianhänge

PEPPOL-EN16931-UBL.xsl	232 KB	2026-03-10	Benjamin DIELEMAN